# MAS508-G Control Theory
# Lecture 4: Stability

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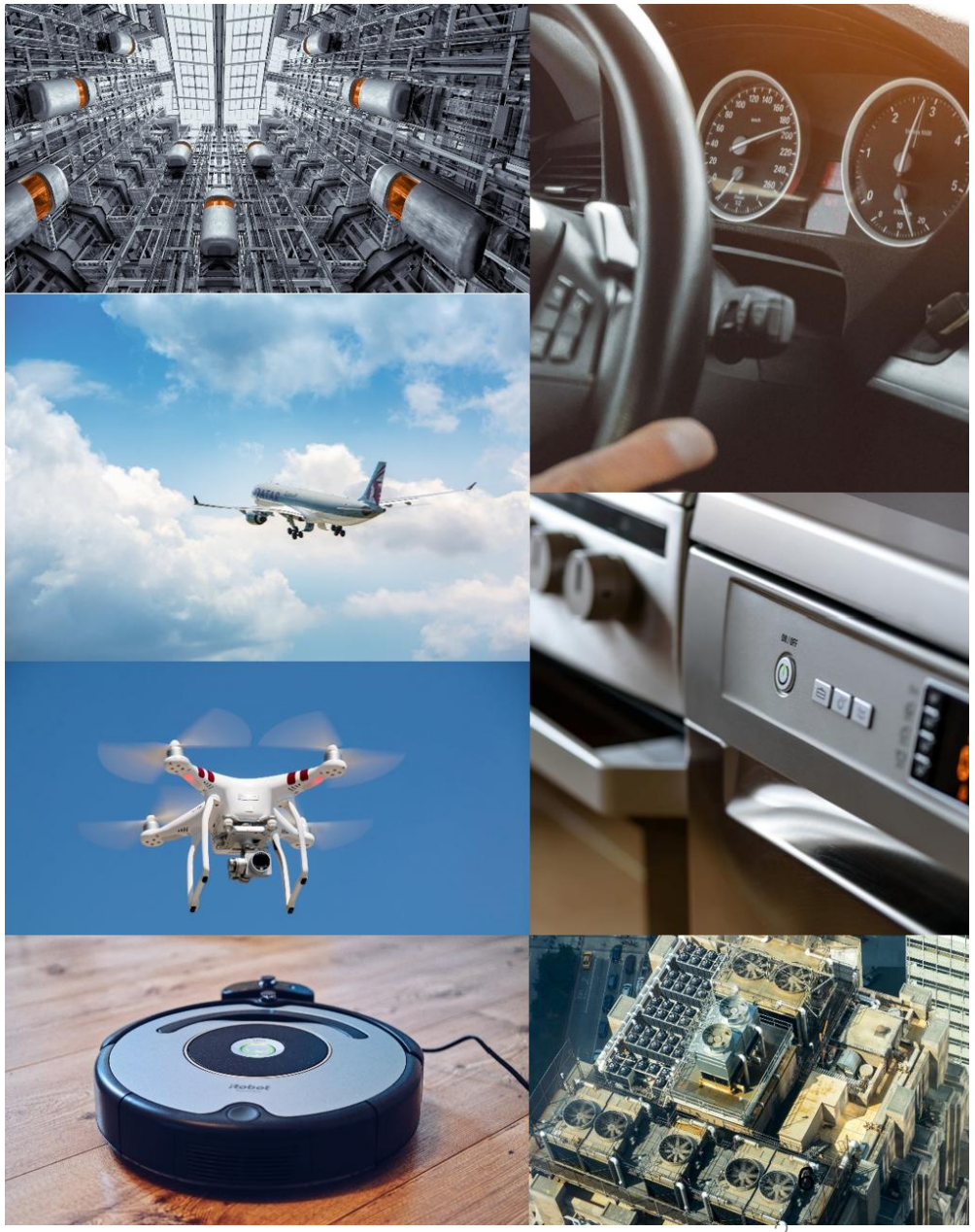
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# Don’t Lose Control!

- Stability is the process of determining if (and when) the state of a system can exponentially increase
- There are various types of stability, and ways that control systems can be used to prevent a system from becoming unstable



# Stability of Linear Time Invariant (LTI) System

- The response of a linear time invariant system can be represented as the sum of the *natural response* and the *forced response*

- The formal definition of stability for a linear time-invariant system based on its natural response is
  1. A LTI system is **stable** if the natural response goes to zero as time goes to infinity
  2. A LTI system is **unstable** if the natural response eventually goes to infinity as time goes to infinity
  3. A LTI system is **marginally stable** if the natural response neither decays nor grows but remains constant or oscillates as time approaches infinity.

# BIBO Stability

The most basic stability metric used in control is **Bounded -Input Bounded-Output Stability** which is defined as:

- A system is stable if every bounded input yields a bounded output

- A system is unstable if any bounded input yields an unbounded output



# BIBO Stability

- The **BIBO stability** of a system is often determined by **observing the response of the system experimentally**, though if the system model is known it can be determined using the final value theorem.

- It is **not** a good idea to **test the stability of a system that might be unstable** if it could result in damage to the system or users.

- **Assessing the BIBO stability of a system, especially marginal stability, can be impractical** depending on the type of system and range of potential inputs.

# Takeaways

- BIBO stability states that
  - ✓ **A system is stable if every bounded input yields a bounded output**
  - ✓ **A system is unstable if any bounded input yields an unbounded output**
  - ✓ **A system is unstable if there is an input that causes the forced response to go to infinity.**

- Otherwise, the system is stable if the natural response approaches zero, or marginally stable if the natural response is constant or oscillates.

- Though conceptually BIBO stability is simple, assessment typically involves testing a system which can be difficult and time consuming

# Stability

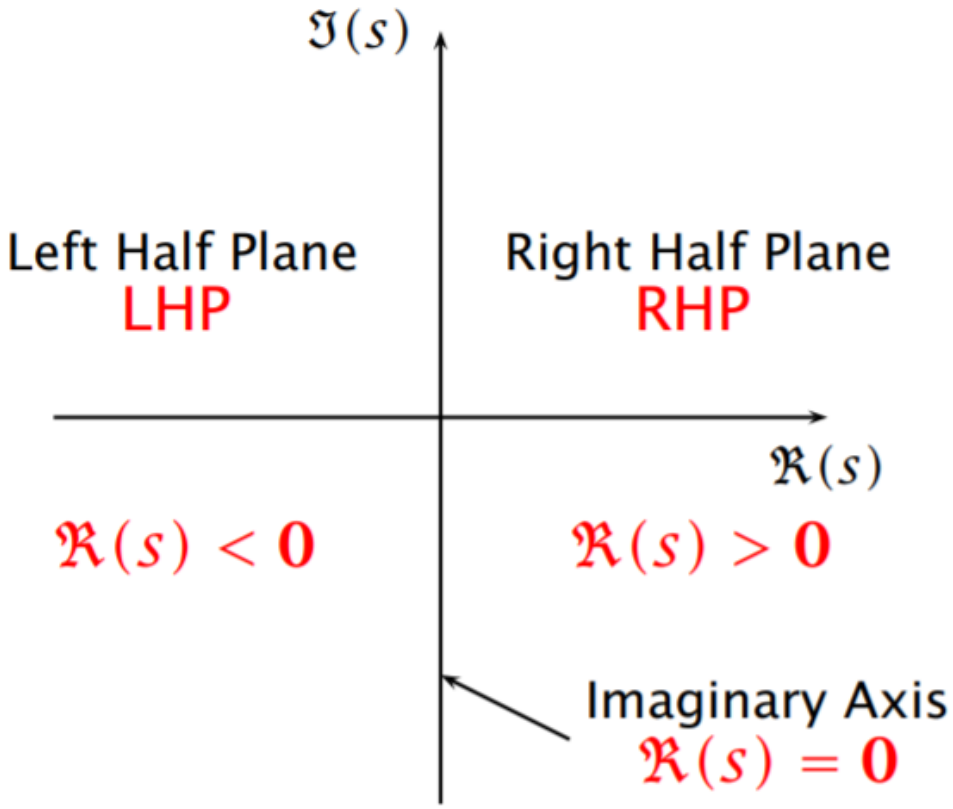
Techniques:

- Poles
- Frequency response technique
- Lyapunov stability
- Routh-Hurwitz

# Pole Stability

- Assessing the stability of a building is a critical part of the design process.

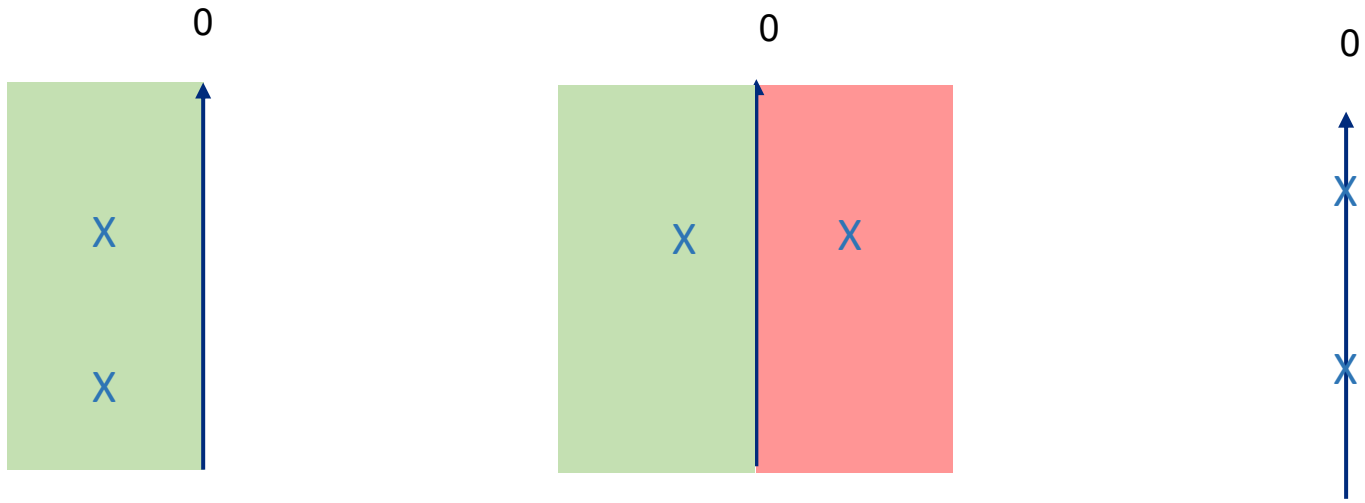
- If a mathematical model of a linear time-invariant system is known, the stability of the system can be determined using **the location of the poles of the system.**



# Stability from Poles

The definition of stability based on pole locations is:

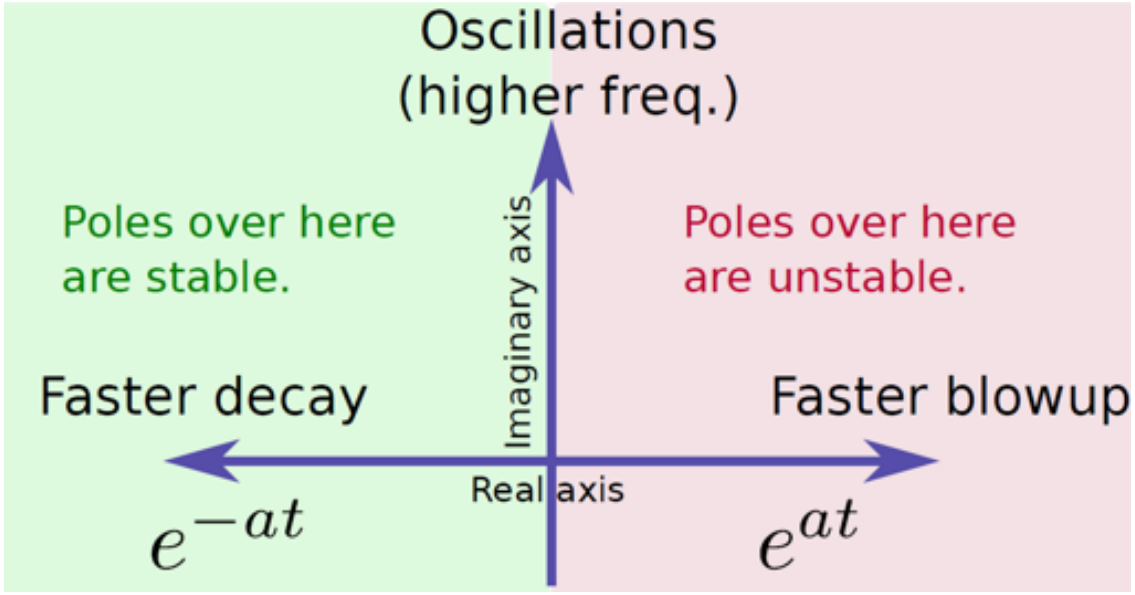
1. The system is **stable** if all its poles are in the **left-hand plane**
2. The system is **unstable** if any poles are in the **right-hand plane** and/or any ***repeated*** **poles** are on the imaginary axis
3. The system is **marginally stable** if it **has *non-repeated* poles on the imaginary axis, and other poles in the left-hand plane.**



# Stability from Poles

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# Examples

**Stable:**

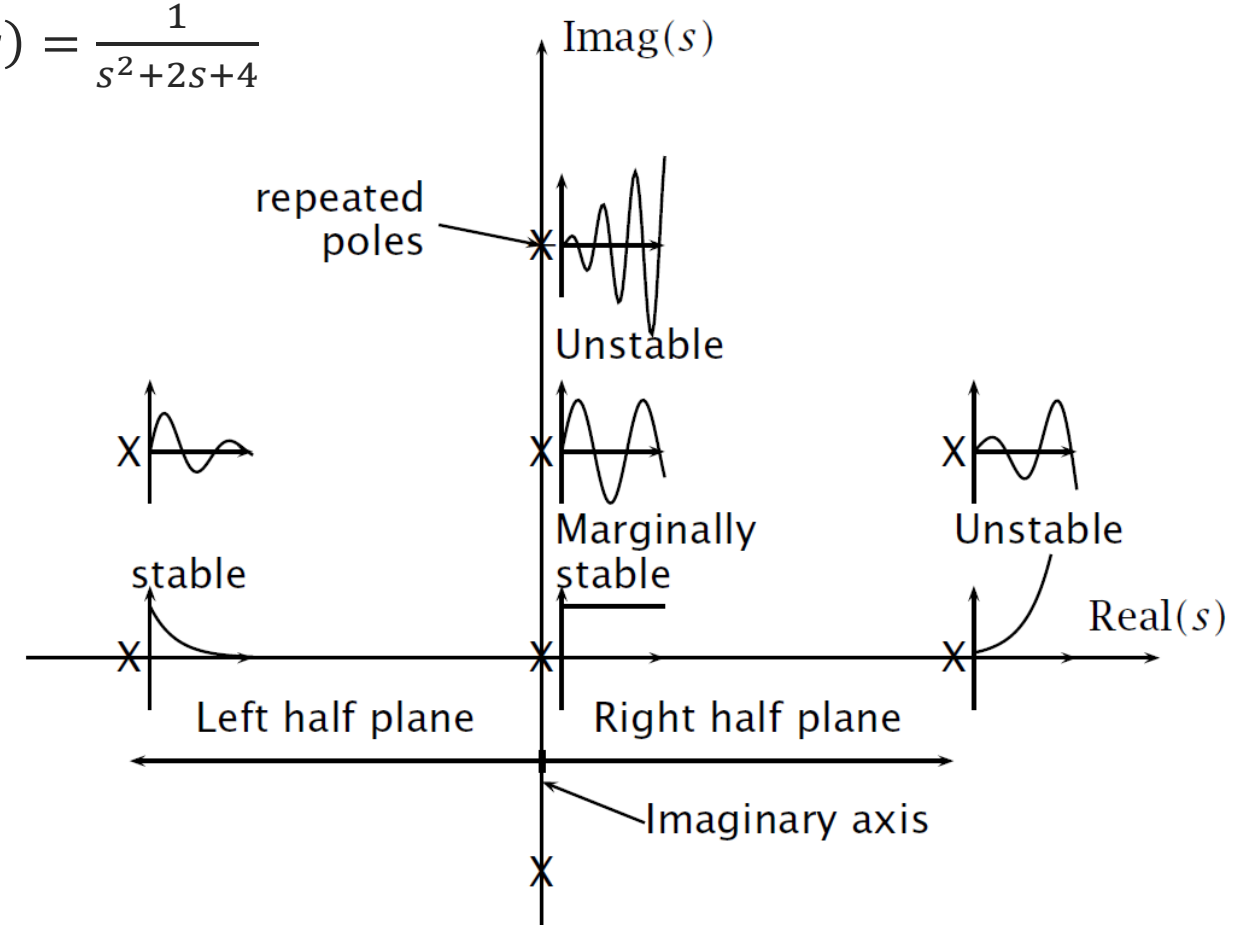
- Underdamped mass-spring-damper system $G(s) = \frac{1}{s^2+2s+4}$

**Marginally stable:**

- Integrator $G(s) = \frac{1}{s}$
- Undamped mass-spring system $G(s) = \frac{1}{s^2+4}$

**Unstable**

- Inverted pendulum $G(s) = \frac{s}{(s-4)(s+4)}$
- Two integrators in series $G(s) = \frac{1}{s^2}$
- Repeated poles $G(s) = \frac{s}{(s^2+4)^2}$



# Takeaways

- Pole Stability can be used to assess the stability of linear time-invariant systems when a model of the system is known.
- Care must be paid to the difference between marginally stable, and unstable systems as it can depend on the *number* of repeated poles are on the imaginary axis.
- More complex or non-linear systems require an alternative approach.

# Stability in State Space

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

Take Laplace transform for the state space model:

$$SX(s) = AX(s) + BU(s)$$
$$Y(s) = CX(s) + DU(s)$$

Solving:

$$X(s) = (sI - A)^{-1}BU(s)$$

$$Y(s) = [\, C(sI - A)^{-1}B + D \,]U(s)$$

If it is SISO system

$$T(s) = \frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D$$

# Pole Stability in State Space

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$$y = Cx + Du$$

If it is SISO system

$$T(s) = \frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D$$

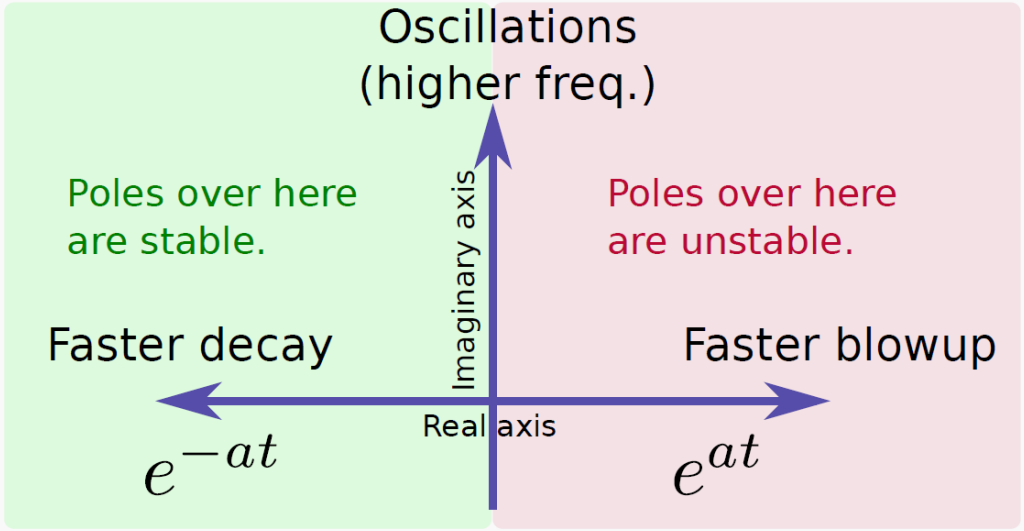
The system transfer function has $\boldsymbol{det(sI - A)}$ in the denominator. Thus

$$\mathbf{det}(\boldsymbol{sI - A})=0$$

is the characteristic equation for the system from which the system poles can be found. (The eigenvalue of A are the poles of the system.)

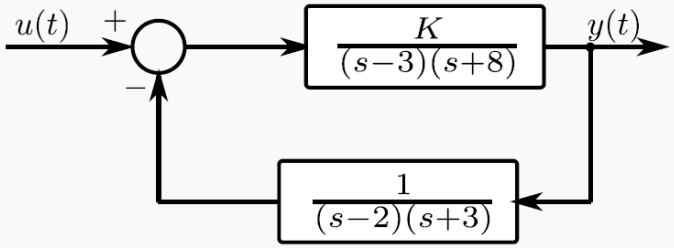
# Frequency response technique (Bode plots)

# Frequency response technique



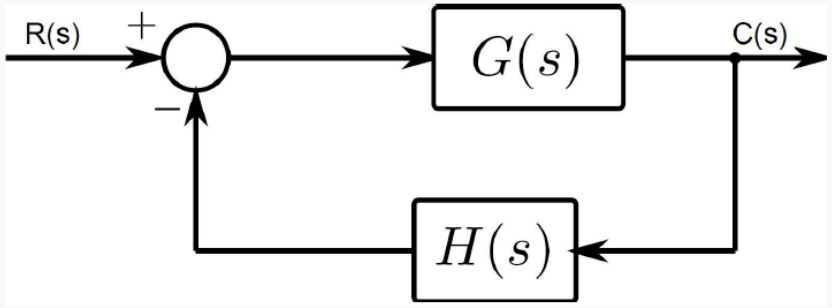

If a system has any unstable poles it is unstable

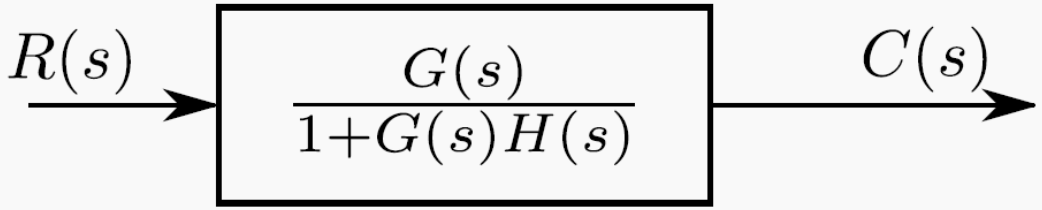
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Is this system stable?

# Stability equivalence



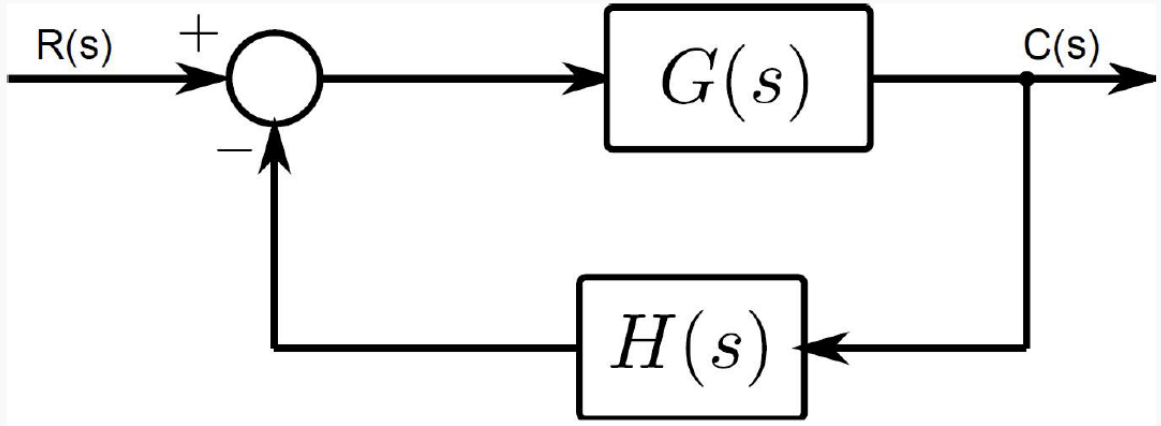




Closed-loop TF: $T(s) = \frac{G(s)}{1+G(s)H(s)}$

Closed-loop system is stable $\Leftrightarrow$ Number of poles of $T(s)$ in RHP $= 0$

# Stability via Bode plots

Determine the stability of the closed-loop system

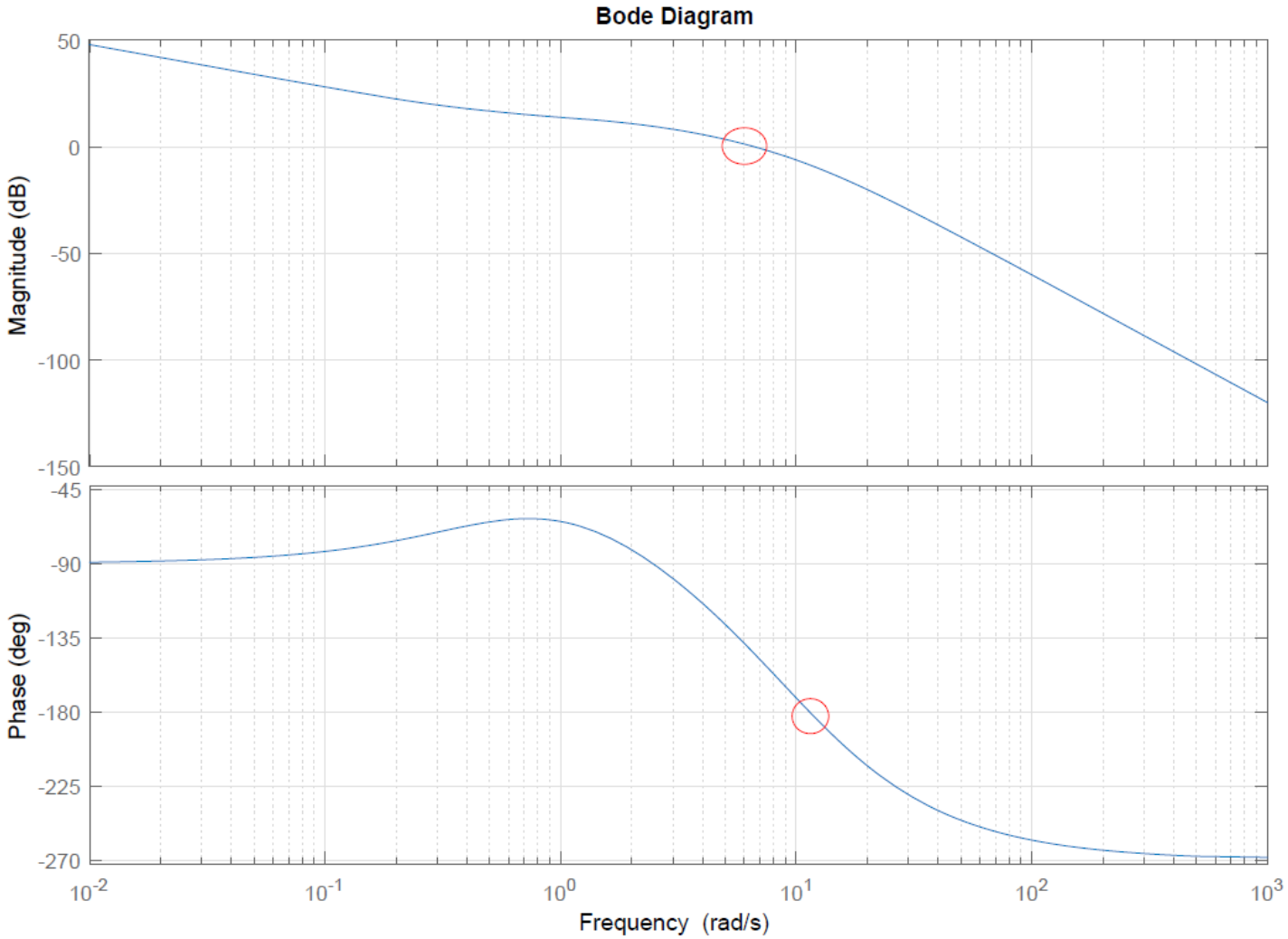



**Bode plot of Open-loop Transfer Function $G(s)H(s)$**

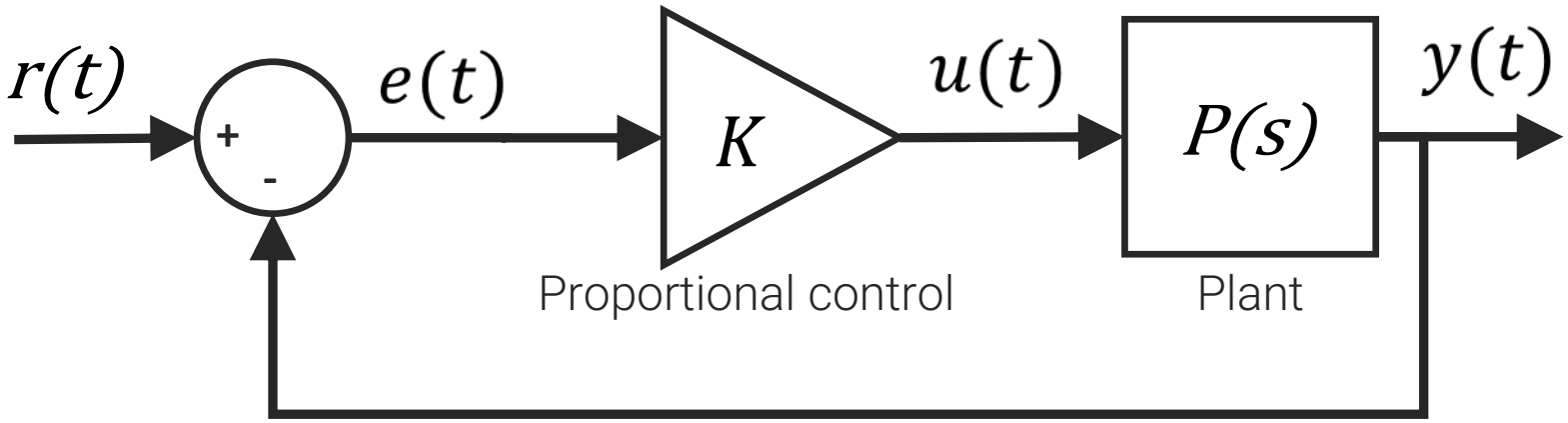
The magnitude is less than $0$ dB (unity gain) at the frequency where the phase is $\pm 180^o$.

$\Rightarrow$ The closed-loop system is Stable.

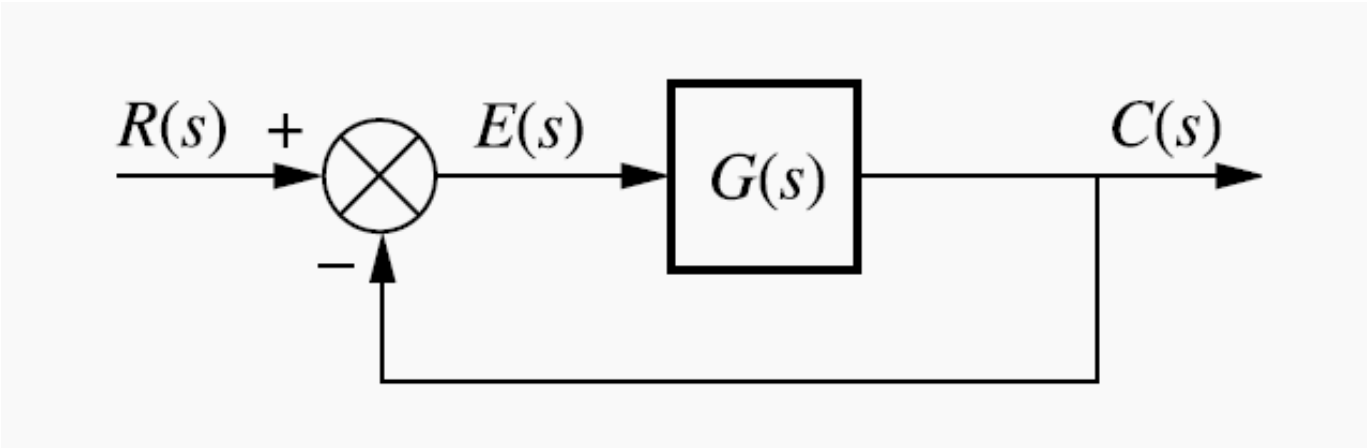
# Example: Stability via bode plots



# Example: P control for a system $P(s)$



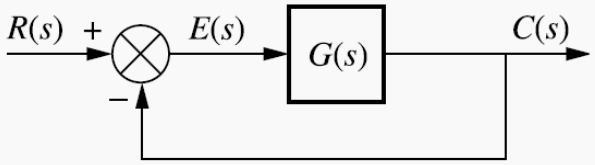

$$P(s) = \frac{1}{s(s+2)(s+4}$$



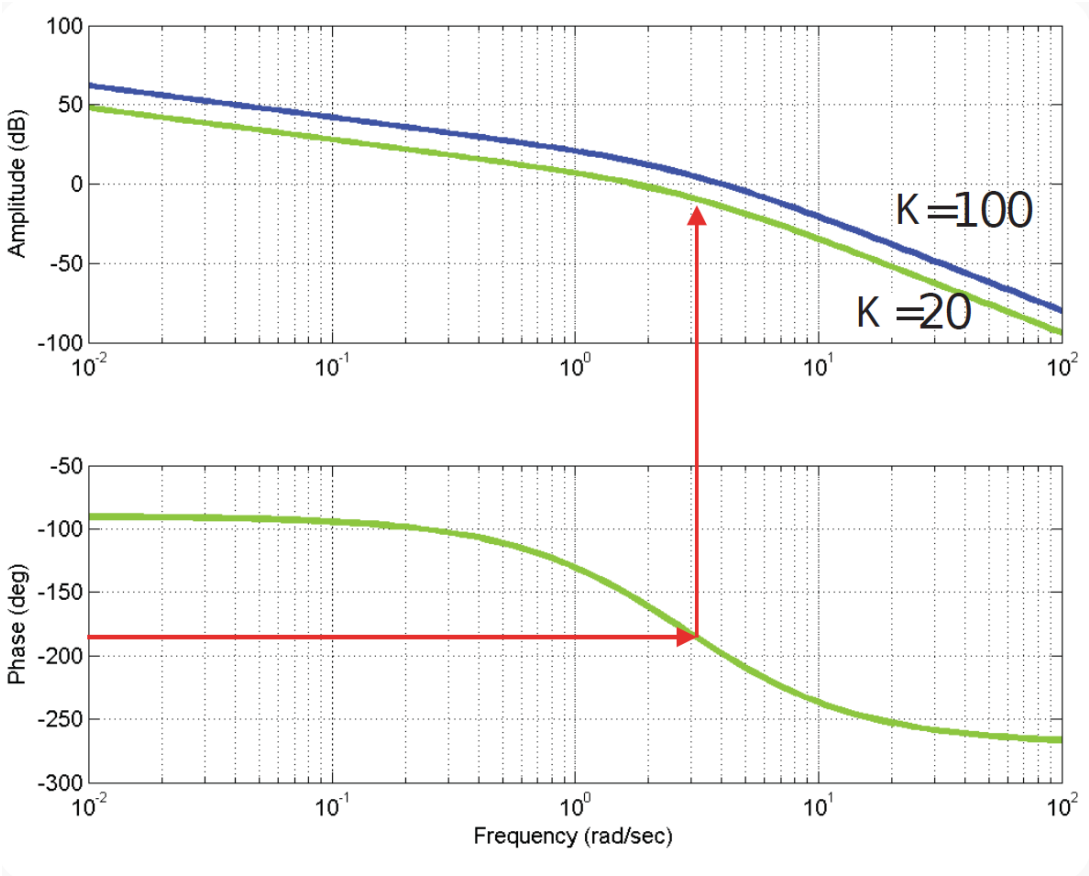

A unity feedback system

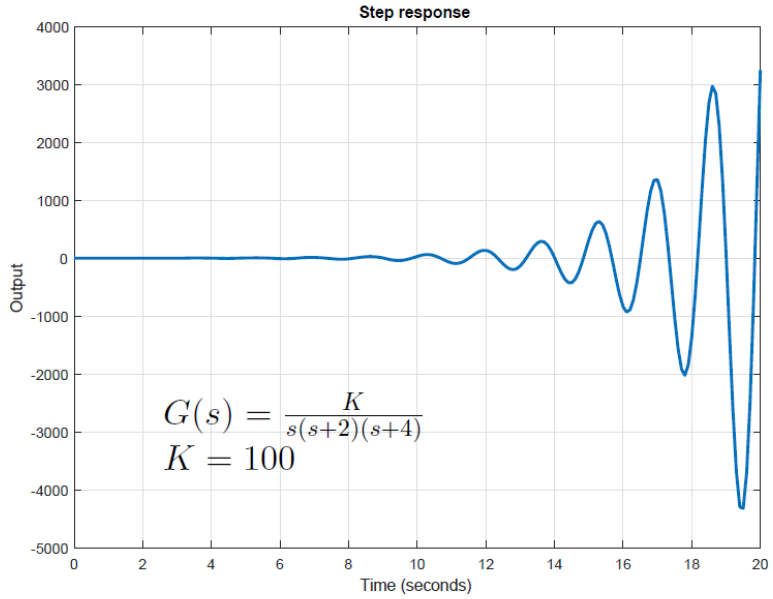
$$G(s) = \frac{K}{s(s+2)(s+4}$$

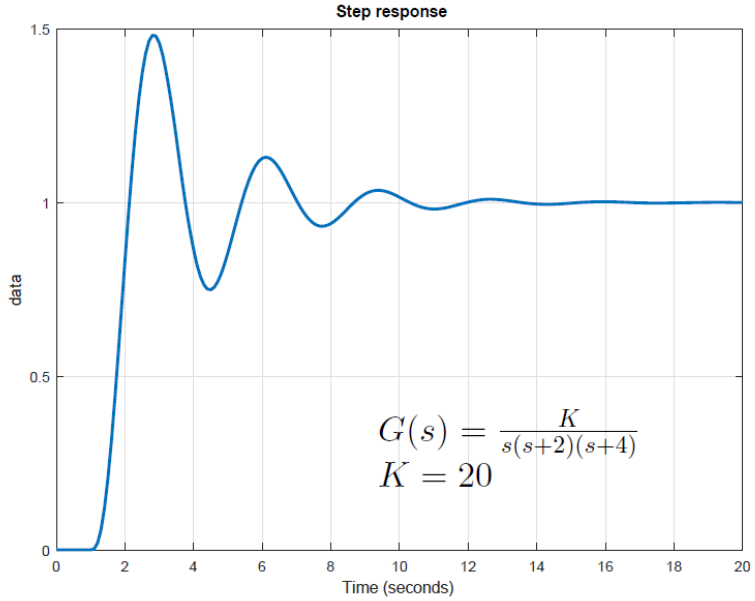
# Example




A unity feedback system with $G(s) = \dfrac{K}{s(s+2)(s+4}$

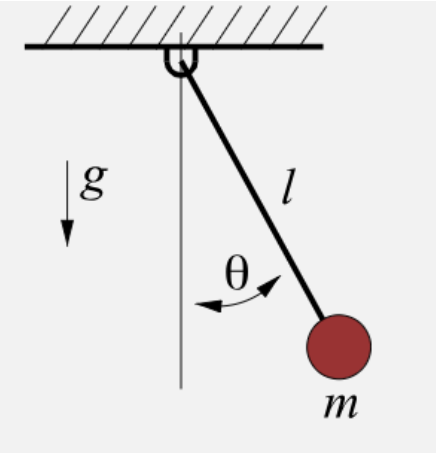


# Lyapunov stability

# Example: Pendulum system




Nonlinear term in equation

Nonlinear equation of motion for this system is:

$$ml^2\ddot{\theta} + c\dot{\theta} + mgl\sin\theta = u$$

Where: *m* is the mass of the pendulum, *c* is the viscous damping, *g* is the gravitational constant, *l* is the distance to the center of mass, and *u* is the applied torque at the pivot.

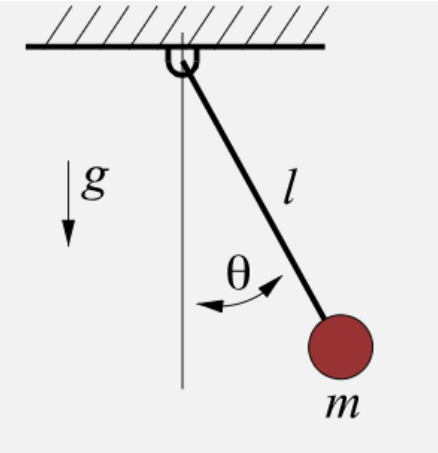
Linearize the nonlinear function $\sin(\theta)$ at operating point $\theta := \theta_0 = 0$ and $u := u_0 = 0$:

$$\boldsymbol{sin(\theta) \approx sin(\theta_0) + cos(\theta_0)(\theta - \theta_0) = \theta}$$

The linearization equation is

$$\boldsymbol{ml^2\,\ddot{\theta} + c\,\dot{\theta} + mgl\,\theta = u}$$

# Example: Pendulum system




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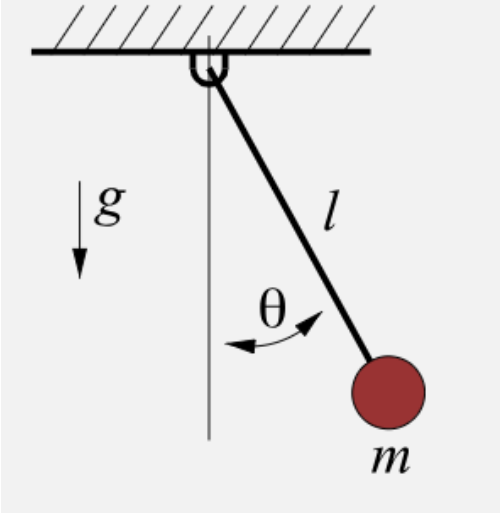
If there is no external control $\boldsymbol{u = 0}$, then

$$\boldsymbol{ml^2\,\ddot{\theta} + c\,\dot{\theta} + mgl\,sin\,\theta = 0}$$

The total system energy (kinetic + potential), which we can write down:

$$\boldsymbol{E(\theta,\dot{\theta}) = \frac{1}{2}ml^2\,\dot{\theta}^2 - mgl\,cos\,\theta}$$

# Example: Stability of Pendulum




Pendulum system

$$\boldsymbol{ml^2\,\ddot{\theta} + c\,\dot{\theta} + mgl\,sin\,\theta = 0}$$

The total system energy (kinetic + potential), which we can write down:

$$\boldsymbol{E(\theta,\dot{\theta}) = \frac{1}{2}ml^2\,\dot{\theta}^2 - mgl\,cos\,\theta \geq 0}$$

A natural route to proving the stability of the downward fixed points is by arguing that energy (almost always) decreases for the damped pendulum ($c > 0$) and so the system will eventually come to rest at the minimum energy, $\boldsymbol{E = -mgl}$, which happens at $\boldsymbol{\theta = 2\pi k}$ **.**

Another way: Evaluating the time derivative of the energy reveals

$$\boldsymbol{\dot{E} = \frac{\partial E}{\partial \dot{\theta}}\,\ddot{\theta} + \frac{\partial E}{\partial \theta}\dot{\theta} = ml^2\dot{\theta}\,\ddot{\theta} + mgl\,sin\theta\,\dot{\theta} = \left(ml^2\,\ddot{\theta} + mgl\,sin\theta\right)\dot{\theta} = -c\dot{\theta}^2 \leq 0}$$

So, we know that energy will never increase, that energy is bounded, and that it obtains its minima at the downward (stable) fixed points.

# Main idea of Lyapunov stability

**Lyapunov formalized the idea:**

*If the total energy is dissipated, then the system must be stable.*

**Main benefit:** By looking at how an energy-like function $V$ (so called Lyapunov function) changes over time, we might conclude that a system is stable or asymptotically stable without solving the nonlinear differential equation.

**Analysis:** Check if $V$ is decreasing with time:

- Continuous time: $\frac{dV}{dt} < 0$
- Discrete time: $V(k+1) - V(k) < 0$

**Main question:** **How to find** a Lyapunov function?

# Stability Definition

An equilibrium point $x^*$ of $\dot{x} = f(x)$ (i.e., $f(x^*) = 0$) is

- **locally stable**, if for every $R > 0$ there exists $r > 0$, such that

$$\|x(0) - x^*\| < r \quad \Rightarrow \quad \|x(t) - x^*\| < R, \quad t \geq 0$$

- **locally asymptotically stable**, if locally stable and

$$\|x(0) - x^*\| < r \quad \Rightarrow \quad \lim_{t \to \infty} x(t) = x^*$$

- **globally asymptotically stable**, if asymptotically stable for all $x(0) \in \mathbb{R}^n$.

# Lyapunov Stability Theorem

**Theorem** Let $\dot{x} = f(x), f(x^*) = 0$ where $x^*$ is in the interior of $\Omega \subset \mathbb{R}^n$.

Assume that $V(x)$ is a scalar continuously-differentiable positive definite function, $V : \Omega \to \mathbb{R}$ *is a* $C^1$. If

(1) $V(x^*) = 0$

(2) $V(x) > 0$, for all $x \in \Omega$, $x \neq x^*$

(3) $\dot{V}(x) \leq 0$ along all trajectories of **the system** in $\Omega$

then $x^*$ is locally stable.

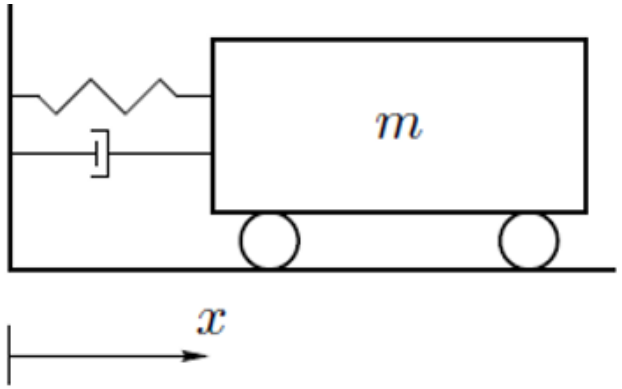
(4) If also $\dot{V}(x) < 0$ for all $x \in \Omega$, $x \neq x^*$

then $x^*$ is locally asymptotically stable.

(5) Furthermore, if $\Omega = \mathbb{R}^n$ and also $V(x) \to \infty$ as $\|x\| \to \infty$

then $x^*$ is globally asymptotically stable.

# Example




$$m\ddot{x} = -\underbrace{b\dot{x}|\dot{x}|}_{\text{damping}} - \underbrace{k_0 x - k_1 x^3}_{\text{spring}}$$

$$b, k_0, k_1 > 0$$

Total energy = kinetic + pot. energy: $V = \frac{mv^2}{2} + \int_0^x F_{spring}\, ds \Rightarrow$

$$V(x, \dot{x}) = m\dot{x}^2/2 + k_0 x^2/2 + k_1 x^4/4 > 0, \qquad V(0,0) = 0$$

$$\frac{d}{dt}V(x, \dot{x}) = m\ddot{x}\dot{x} + k_0 x\dot{x} + k_1 x^3 \dot{x} = \{\text{plug in system dynamics}^2\}$$

$$= -b|\dot{x}|^3 \quad < 0, \text{ for } \dot{x} \neq 0$$

# Lyapunov Stability for Linear Systems

**Linear system:** $\dot{x} = Ax$

*Lyapunov equation:* Let $Q = Q^T > 0$. Solve

$$PA + A^T P = -Q$$

with respect to the symmetric matrix $P$.

*Lyapunov function:* $V(x) = x^T Px$, $\Rightarrow$

$$\dot{V}(x) = x^T P\dot{x} + \dot{x}^T Px = x^T(PA + A^T P)x = -x^T Qx < 0$$

**Asymptotic Stability:** If $P = P^T > 0$, then the Lyapunov Stability Theorem implies (local=global) asymptotic stability, hence the eigenvalues of $A$ must satisfy $\mathrm{Re}\,\lambda_k(A) < 0,\ \forall k$

# Example – Lyapunov stability for linear system

$$\dot{x} = Ax = \begin{bmatrix} -1 & 4 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \qquad (1)$$

Eigenvalues of A : $\{-1, -3\} \Rightarrow$ (global) asymptotic stability.

*Lyapunov function:* $V(x) = x^T P x, \Rightarrow$

$$\dot{V}(x) = x^T P\dot{x} + \dot{x}^T P x = x^T (PA + A^T P)x = -x^T Q x < 0$$

Find a quadratic Lyapunov function for the system (1):

$$V(x) = x^T P x = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad P = P^T > 0$$

Take any $Q = Q^T > 0$ , say $Q = I_{2\times 2}$. Solve $A^T P + PA = -Q$.

# Example – Lyapunov stability for linear systems

$$A^T P + PA = -I$$

$$\begin{bmatrix} -1 & 0 \\ 4 & -3 \end{bmatrix} \begin{bmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{bmatrix} + \begin{bmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{bmatrix} \begin{bmatrix} -1 & 4 \\ 0 & -3 \end{bmatrix} =$$
$$\begin{bmatrix} -2p_{11} & -4p_{12} + 4p_{11} \\ -4p_{12} + 4p_{11} & 8p_{12} - 6p_{22} \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

Solving for $p_{11}$, $p_{12}$ and $p_{22}$ gives

$$\begin{aligned} 2p_{11} &= -1 \\ -4p_{12} + 4p_{11} &= 0 \\ 8p_{12} - 6p_{22} &= -1 \end{aligned}$$

$$\Longrightarrow \begin{bmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 5/6 \end{bmatrix} > 0$$

# Lyapunov-Based Control Design

For a system $\dot{x} = f(x, u)$

Idea: Shape $\dot{V}(x)$ via control input

Procedure:

- Choose Lyapunov function $V(x)$ as a continuous differentiable positive definite function
- Compute $\dot{V}(x)$ with system dynamics
- Design control $u(x)$ to make $\dot{V}(x) \leq -W(x)$ , where $W(x)$ is positive definite.

# Example 1: Mass–Spring–Damper

Plant: $m\,\ddot{x} + b\,\dot{x} + k\,x = u$

State: $x_1 = x, x_2 = \dot{x}$

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = \frac{1}{m}(-bx_2 - kx_1 + u).$$

Lyapunov function $V = \frac{1}{2}m\,x_2^2 + \frac{1}{2}k\,x_1^2$

$$\dot{V} = mx_2\dot{x}_2 + kx_1\dot{x}_1 = x_2\left(-bx_2 - kx_1 + u\right) + kx_1x_2 = -bx_2^2 + x_2u.$$

Control law: $u = -k_d\,x_2$ (damping injection, $k_d > 0$ )

$$\dot{V} = (-k_d + b)x_2^2$$

# Example 1: Mass–Spring–Damper

$$\dot{V} = (-k_d + b)x_2^2$$

So $\dot{V}$ is negative semi-definite (zero when $x_2 = 0$).

**Remarks on stability.**

- $V$ is positive definite. $\dot{V} \leq 0$ means the origin is stable (Lyapunov).
- To conclude **asymptotic stability**, use **LaSalle's invariance principle**: the set where $\dot{V} = 0$ is $\{x \mid x_2 = 0\}$. On that set, the dynamics give $\dot{x}_1 = x_2 = 0$ and $\dot{x}_2 = -(k/m)x_1$. For invariance, we must have $\dot{x}_2 = 0$ too, thus $x_1 = 0$. Hence the only invariant set within $\{x_2 = 0\}$ is the origin. Therefore all trajectories converge to the origin: the origin is **globally asymptotically stable**.

# Example 2: nonlinear system

Consider the nonlinear system

$$\dot{x}_1 = -3x_1 + 2x_1x_2^2 + u$$
$$\dot{x}_2 = -x_2^3 - x_2,$$

Find a nonlinear feedback control law which makes the origin globally asymptotically stable.

We try the standard Lyapunov function **candidate**

$$V(x_1, x_2) = \frac{1}{2}\left(x_1^2 + x_2^2\right),$$

which satisfies

- $V(0,0) = 0$
- $V(x_1, x_2) > 0, \quad \forall (x_1,\, x_2) \neq (0,\, 0).$
- radially unbounded,
- compute

$$\dot{V} = \dot{x}_1x_1 + \dot{x}_2x_2 = (-3x_1 + 2x_1x_2^2 + u)x_1 + (-x_2^3 - x_2)x_2$$
$$= -3x_1^2 - x_2^2 + ux_1 + 2x_1^2x_2^2 - x_2^4$$

We would like to have

$$\dot{V} < 0 \qquad \forall (x_1,\, x_2) \neq (0,\, 0)$$

Inserting the control law, $u = -2x_1x_2^2$, we get

$$\dot{V} = -3x_1^2 - x_2^2 \underbrace{-2x_1^2x_2^2 + 2x_1^2x_2^2}_{=0} - x_2^4 = -3x_1^2 - x_2^2 - x_2^4 < 0, \quad \forall x \neq 0$$

According to the LaSalle's Invariance Theorem, the origin globally asymptotically stable is achieved, such as $\lim_{t\to\infty} x_1 = 0, \lim_{t\to\infty} x_2 = 0$

# Routh-Hurwitz

- The Routh-Hurwitz approach to stability allows you to test complex systems analytically without extensive mathematical analysis
- Stability can be assessed without having to know exact pole locations

# Routh-Hurwitz

The Routh-Hurwitz Criterion is a two-step process

1. Create a *Routh table* based on the coefficients of the denominator of the transfer function.
2. Use the table to find the number of poles in the left-hand plane, on the imaginary axis, or in the right-hand plane.

The major benefit of this approach is the ability to determine the ranges for unknown system parameters where the closed-loop system is stable.

# Routh Tables

A Routh table is created by entering the coefficients of the denominator of the transfer function as follows:

For an nth-degree denominator $D(s) = a_n s^n + a_{n-1} s^{n-1} + \ldots + a_1 s + a_0$

| | | | |
|---|---|---|---|
| $a_n$ | $a_{n-2}$ | $a_{n-4}$ | $\cdots$ |
| $a_{n-1}$ | $a_{n-3}$ | $a_{n-5}$ | $\cdots$ |
| $b_1$ | $b_2$ | $b_3$ | $\cdots$ |
| $c_1$ | $c_2$ | $c_3$ | $\cdots$ |
| $\vdots$ | $\vdots$ | $\vdots$ | $\ddots$ |

where $b_i$ and $c_i$ are computed as the negative determinant of a 2 x 2 matrix divided by the value in the first column of the row above. **The number of *sign changes* in the first column corresponds to the number of poles in the right-hand plane.**

# Example of 4th order

| | | | |
|---|---|---|---|
| $s^4$ | $a_4$ | $a_2$ | $a_0$ |
| $s^3$ | $a_3$ | $a_1$ | $0$ |
| $s^2$ | $\dfrac{-\begin{vmatrix} a_4 & a_2 \\ a_3 & a_1 \end{vmatrix}}{a_3} = b_1$ | $\dfrac{-\begin{vmatrix} a_4 & a_0 \\ a_3 & 0 \end{vmatrix}}{a_3} = b_2$ | $\dfrac{-\begin{vmatrix} a_4 & 0 \\ a_3 & 0 \end{vmatrix}}{a_3} = 0$ |
| $s^1$ | $\dfrac{-\begin{vmatrix} a_3 & a_1 \\ b_1 & b_2 \end{vmatrix}}{b_1} = c_1$ | $\dfrac{-\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix}}{b_1} = 0$ | $\dfrac{-\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix}}{b_1} = 0$ |
| $s^0$ | $\dfrac{-\begin{vmatrix} b_1 & b_2 \\ c_1 & 0 \end{vmatrix}}{c_1} = d_1$ | $\dfrac{-\begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix}}{c_1} = 0$ | $\dfrac{-\begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix}}{c_1} = 0$ |

# Example: Routh Tables

For the denominator $D(s) = s^3 + 2s^2 + 3s + 5$

| | |
|---|---|
| 1 | 3 |
| 2 | 5 |
| $\frac{-\begin{vmatrix} 1 & 3 \\ 2 & 5 \end{vmatrix}}{2} = -\frac{1}{2}$ | 0 |
| 5 | 0 |

Since there are two sign changes in the first column ([2→-1/2],[-1/2→5]), the system is unstable.

# Routh Table Special Cases

- First column value in any row is 0
  - Replace the 0 with a small number, $\epsilon$. Continue finding the other entries in the table and determine if a sign change occurs when $\epsilon$ approaches zero.
- Entire row is 0
  - Take the coefficients of the row above and construct a polynomial P(s)
  - Differentiate the polynomial and use its coefficients in the row of zeros

# Takeaways

- The Routh-Hurwitz criterion is used to assess the stability of complex systems analytically without needing to know the specific location of the poles.
- The denominator coefficients are used to populate a table where the number of changes in sign of the first column determines the stability of the system.